

Agentic AI, Vertex AI & Generative AI Interview Question Bank

1. Generative AI Fundamentals

1. What is Generative AI?
 2. How does ChatGPT work?
 3. Explain the architecture of a Large Language Model (LLM).
 4. What are tokens, embeddings, context windows, and model parameters?
 5. What are the limitations of LLMs?
 6. What causes hallucinations in LLMs?
 7. How do you reduce hallucinations?
 8. What are the different methods to improve response accuracy?
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2. Agentic AI & Multi-Agent Systems

9. What is Agentic AI?
 10. How is Agentic AI different from a traditional LLM application?
 11. LLM vs Agentic AI – explain the differences.
 12. What is the architecture of an AI Agent?
 13. What are the key components of an Agentic AI system?
 14. What is a Single-Agent system?
 15. What is a Multi-Agent system?
 16. When would you choose a Multi-Agent system instead of a Single-Agent system?
 17. Why do we need multiple agents when one agent can perform multiple tasks?
 18. What are the benefits and challenges of Multi-Agent systems?
 19. What is an A2A (Agent-to-Agent) agent?
 20. What is a Sequential Agent?
 21. What is a Loop Agent?
 22. Difference between Sequential Agents and Loop Agents.
 23. How do agents communicate with each other?
 24. Explain an end-to-end Agentic AI solution you have built.
 25. Describe the architecture, component interaction, and business problem solved.
 26. How do you monitor Agentic AI applications in production?
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3. Agent Frameworks

27. What is ADK (Agent Development Kit)?

- 28. What is TDK Agent Development?
 - 29. What are the benefits of ADK?
 - 30. ADK vs LangChain.
 - 31. ADK vs CrewAI vs AutoGen.
 - 32. What is LangGraph?
 - 33. What are LangGraph frameworks?
 - 34. When would you use LangGraph over LangChain?
 - 35. How do you build multi-agent workflows using LangGraph?
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4. MCP (Model Context Protocol)

- 36. What is MCP (Model Context Protocol)?
 - 37. Why is MCP required?
 - 38. Have you used MCP in your projects?
 - 39. Can you explain a use case where you implemented MCP?
 - 40. How does an MCP server work?
 - 41. Can you provide an example architecture involving MCP?
 - 42. MCP vs traditional API integrations.
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5. Vertex AI & Google Cloud

- 43. What services are available in Vertex AI?
 - 44. What can be achieved using Vertex AI services?
 - 45. Have you worked with Vertex AI?
 - 46. What is Vertex AI Model Garden?
 - 47. What is Vertex AI Agent Builder?
 - 48. What is Vertex AI Search?
 - 49. What is Vertex AI Vector Search?
 - 50. How do you deploy a Generative AI application on Vertex AI?
 - 51. How do you monitor AI applications on Vertex AI?
 - 52. How do you evaluate Gemini models?
 - 53. How do you deploy an Agentic AI application on GCP?
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6. Grounding & Hallucination Reduction

- 54. What is grounding in Vertex AI?
- 55. Why is grounding required?
- 56. What are grounding techniques?
- 57. Grounding vs RAG.
- 58. What are different approaches to grounding?

- 59. How do grounding techniques reduce hallucinations?
 - 60. What other methods can reduce hallucinations besides grounding?
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7. RAG (Retrieval-Augmented Generation)

- 61. What is RAG?
 - 62. What does RAG stand for?
 - 63. Explain an end-to-end RAG pipeline.
 - 64. What are the components of a RAG system?
 - 65. How does retrieval work in RAG?
 - 66. What are embeddings in RAG?
 - 67. How do vector embeddings work?
 - 68. What is semantic search?
 - 69. Difference between semantic search and keyword search.
 - 70. What are common challenges in RAG implementations?
 - 71. How do you troubleshoot RAG systems?
 - 72. What causes latency in RAG systems?
 - 73. How do you reduce RAG latency?
 - 74. How do you evaluate a RAG system?
 - 75. Which metrics do you use to evaluate RAG quality?
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8. Chunking & Vector Databases

- 76. What is chunking?
- 77. Why is chunking important in RAG?
- 78. What chunking strategies have you used?
- 79. Fixed-size chunking.
- 80. Recursive chunking.
- 81. Semantic chunking.
- 82. Token-based chunking.
- 83. Sentence-based chunking.
- 84. Hierarchical chunking.
- 85. What is chunk overlap?
- 86. How do you determine the optimal chunk size?

Vector Databases

- 87. What are vector databases?
- 88. Which vector databases have you worked with?
- 89. Compare Pinecone, Weaviate, ChromaDB, FAISS, and Vertex AI Vector Search.
- 90. How does vector indexing work?
- 91. How does similarity search work?

9. Prompt Engineering

- 92. What is Prompt Engineering?
- 93. Why is Prompt Engineering important?
- 94. What is prompt authenticity?
- 95. What are system prompts and user prompts?
- 96. What is Zero-Shot Prompting?
- 97. What is One-Shot Prompting?
- 98. What is Few-Shot Prompting?
- 99. Difference between Zero-Shot and Few-Shot Prompting.
- 100. What is Chain of Thought (CoT) prompting?
- 101. How do you improve prompt quality?

10. Data Engineering & BigQuery

- 102. What is a data pipeline?
- 103. Explain an end-to-end data ingestion pipeline.
- 104. What is batch data ingestion?
- 105. Batch vs Streaming ingestion.
- 106. What are ETL and ELT?
- 107. What is data partitioning?
- 108. What is clustering in BigQuery?
- 109. Partitioning vs Clustering.
- 110. How do partitioning and clustering improve performance?
- 111. How do you ensure data quality in pipelines?

11. Deployment & MLOps

- 112. Explain the deployment process for a GenAI application.
- 113. How do you deploy applications on GCP?
- 114. What CI/CD practices do you follow?
- 115. How do you monitor AI systems in production?
- 116. How do you perform model versioning?
- 117. What observability tools have you used?
- 118. How do you secure GenAI applications?

12. Python & Coding

119. How would you rate yourself in Python out of 10?
 120. What Python libraries do you use for Generative AI?
 121. How do you call the Gemini API using Python?
 122. Write Python code to initialize and invoke a Gemini model.
 123. How do you handle API errors in Python?
 124. How do you structure production-grade Python applications?
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13. Git & Version Control

125. What is Git?
 126. Explain the Git workflow.
 127. Common Git commands used daily.
 128. Difference between Git Merge and Git Rebase.
 129. How do you resolve merge conflicts?
 130. What is a Pull Request?
 131. What branching strategy do you follow?
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14. Scenario-Based Questions

132. Design an enterprise RAG solution.
133. Design a customer support Agentic AI solution.
134. Design a Multi-Agent workflow using LangGraph.
135. How would you reduce hallucinations in a healthcare chatbot?
136. How would you optimize a slow RAG pipeline?
137. How would you build an end-to-end Agentic AI application on Vertex AI?
138. Explain a challenging GenAI project you have delivered end-to-end.
139. What challenges did you face and how did you solve them?